Ch 05: Value Returning Functions, Python Modules

Application Program Development

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Value-Returning Functions

Value-Returning Function

A value-returning function is a function that returns a value back to the caller. Python, as well as most other programming languages, provide a library of prewritten functions that perform commonly needed tasks.

```
def value_returning_function(parameter1, parameter2, ...):
    calculation1
    calculation2
    return result

r = value_returning_function(p1, p2, ...)
```

- **void function**: group of statements within a program for performing a specific task
  - Call function when you need to perform the task
  - The task is done as a *side-effect* of calling a void function
- **Value-returning function**: In contrast a value-returning function calculates a result from input and returns the result from calling the function
  - Value returned to the caller that invoked the function

Standard Library Functions and the `import` Statement

- **Standard library**: library of pre-written functions that comes with a programming language like Python
  - **Library functions** perform tasks that programmers commonly need
    - Example: `print()`, `input()`, `range()`
    - Viewed by programmers as a "black box"
- Some library functions built into Python interpreter
  - To use, just call the functions
- **Modules**: collections of functions in standard library
  - Help organize library functions not built into the interpreter
  - Module **namespaces**
- To call a function stored in a module, need to write an `import` statement
  - Written at the **top** of the program
  - Format: `import module_name`

Figure 1: A library function viewed as a black box (Gaddis, 2021, pg.273)

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Generating Random Numbers (1 of 2)

- Random numbers are useful in a log of programmint tasks
- **random** module: includes library functions for working with random numbers
- **Dot notation** notation for calling a function in a module namespace
  - Format: `module_name.function_name()`
- **`randint()` function**: generates a random number in the range provide by the arguments
  - Returns the random number ot part of program that called the function
  - Returned integer can be used anywhere that an integer can
  - You can experiment with the function in interactive mode

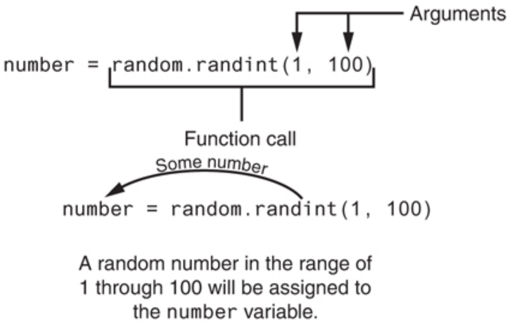

Figure 2: The random function returns a random int in asked for range (Gaddis, 2021, pg.274)

Generating Random Numbers (2 of 2)

- `randrange()` **function**: similar to `range()` function, but returns randomly selected integer from the resulting sequence
  - Same arguments as for the `range()` function
- `random()` **function**: returns a random `float` in the range of 0.0 to 1.0
  - Does not have any arguments
- `uniform()` **function**: returns a random `float` but allows user to specify range

Pseudo-Random Numbers

Random number created by functions in `random` module are actually pseudo-random numbers.

- **Seed value**: initializes the formula that generates random numbers
  - Need to use different seeds in order to get different series of random numbers
    - By default uses system time for seed
    - Can use `random.seed()` function to specify desired seed value

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Writing Your Own Value-Returning Functions

Value-Returning Function

A value-returning function has a `return` statement that returns a value back to the caller of the function.

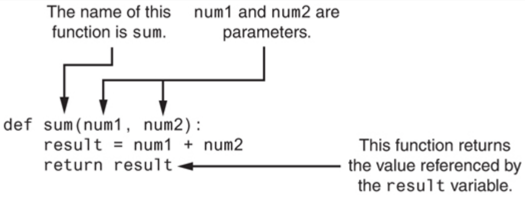

Figure 3: Parts of a function (Gaddis, 2021, pg.283)

- To write a value-returning function, you write a simple function and add one or more `return` statements
  - Format: `return expression`
    - The value for `expression` will be returned to the call of your function
  - The the expression in the `return` statement cane be a complex expression, such as a sum of two variables or the result of another value-returning function

Summary

References

Gaddis, T. (2021). *Starting out with python* (5th). Pearson.